

Electric Charge and Electric Field, **solved**

Step-by-step solutions to every practice problem in Chapter 13. Try each problem on your own before you read its solution.

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P13.1

The charged balloon or comb **polarises** the neutral object. In an insulator such as paper or plaster, each molecule's electron cloud shifts slightly relative to its nuclei. In a conductor, free electrons move. Either way, the side of the object facing the charged body gains a slight excess of the *opposite* charge, and the far side an equal excess of the *same* charge.

The opposite charge is closer, and Coulomb's force falls off as $1/r^2$, so the attraction to the near side beats the repulsion from the far side. The net force is **attractive**.

Opposite charge? Yes, it still works. Reversing the comb's charge reverses the polarisation, and the opposite charge is again on the near side. A charged object attracts a neutral one whatever the sign of its charge.

P13.2

The field at any point has a single, definite direction: the direction of the force on a positive test charge placed there. A field line follows that direction. If two lines crossed, the field at the crossing point would have two different directions at once, which is impossible. The only exception is a point where $\vec{E} = 0$. There the direction is undefined, but such points are not crossings of genuine field lines.

P13.3

Yes, attraction. The positive charge pulls the sphere's free electrons towards the near side, leaving positive charge on the far side. The sphere stays neutral overall. But the negative charge is closer to the external charge than the positive charge is, so the attraction (stronger, at shorter range) exceeds the repulsion. The net force pulls the sphere **towards** the point charge. Any charged object attracts a neutral conductor.

P13.4

Far from a dipole, the two charges are at almost the same distance, and they have equal and opposite strengths. So their fields almost cancel. What survives is the small *difference* between kq/r^2 evaluated at two points a distance d apart. To first order, this difference is

$$d \times \left| \frac{d}{dr} \left(\frac{kq}{r^2} \right) \right| = \frac{2kqd}{r^3}.$$

The cancellation costs one extra power of r . In general, each further level of cancellation (quadrupole, octupole, ...) costs another power. This is why neutral molecules interact only weakly over long distances.

P13.5

Forces on one sphere: its weight mg (down), the tension T along the thread, and the electric repulsion F (horizontal). Equilibrium gives

$$T \cos \theta = mg, \quad T \sin \theta = F \Rightarrow F = mg \tan \theta.$$

$$F = (1.0 \times 10^{-4})(9.8) \tan 10^\circ = 1.73 \times 10^{-4} \text{ N}.$$

Separation: $r = 2L \sin \theta = 2(0.50) \sin 10^\circ = 0.174 \text{ m}.$

Charge:

$$F = \frac{kq^2}{r^2} \Rightarrow q = r \sqrt{\frac{F}{k}} = 0.174 \sqrt{\frac{1.73 \times 10^{-4}}{8.99 \times 10^9}} = 0.174(1.39 \times 10^{-7}) = 2.4 \times 10^{-8} \text{ C} = 24 \text{ nC}.$$

P13.6

(a) **At the centre.** The four equal charges are symmetrically placed, so the fields from opposite corners cancel in pairs: $\vec{E} = \vec{0}$.

(b) **Force on one charge.** Let $F_0 = \frac{kq^2}{a^2} = \frac{(8.99 \times 10^9)(1.0 \times 10^{-6})^2}{0.010} = 0.899 \text{ N}$.

- The two neighbours, each at distance a , push along the two sides with force F_0 . Their resultant points along the diagonal, outward: $2F_0 \cos 45^\circ = \sqrt{2}F_0 = 1.27 \text{ N}$.
- The opposite corner, at distance $a\sqrt{2}$, pushes along the diagonal with $F_0/2 = 0.45 \text{ N}$.

$$F_{\text{net}} = F_0 \left(\sqrt{2} + \frac{1}{2} \right) = 0.899(1.914) = 1.72 \text{ N},$$

directed along the diagonal, away from the centre.

P13.7

Distances. Both charges are $\sqrt{0.05^2 + 0.12^2} = 0.13 \text{ m}$ from the point. Each field has magnitude

$$E_1 = \frac{(8.99 \times 10^9)(5.0 \times 10^{-9})}{0.13^2} = 2.66 \times 10^3 \text{ N/C}.$$

Directions.

- The field of $+5 \text{ nC}$ points *away* from it, along $(0.05, 0.12)/0.13 = (0.385, 0.923)$.
- The field of -5 nC points *towards* it, along $(0.05, -0.12)/0.13 = (0.385, -0.923)$.

The y -components cancel, and the x -components add:

$$E = 2(2.66 \times 10^3)(0.385) = 2.0 \times 10^3 \text{ N/C, in the } +x \text{ direction,}$$

which is parallel to the line from the positive to the negative charge.

P13.8

Electron.

$$a = \frac{eE}{m_e} = \frac{(1.602 \times 10^{-19})(500)}{9.11 \times 10^{-31}} = 8.79 \times 10^{13} \text{ m/s}^2.$$

$$v = \sqrt{2ad} = \sqrt{2(8.79 \times 10^{13})(0.020)} = 1.9 \times 10^6 \text{ m/s}, \quad t = \frac{v}{a} = 2.1 \times 10^{-8} \text{ s}.$$

Proton.

$$a = \frac{eE}{m_p} = 4.79 \times 10^{10} \text{ m/s}^2, \quad v = \sqrt{2(4.79 \times 10^{10})(0.020)} = 4.4 \times 10^4 \text{ m/s}, \quad t = 9.1 \times 10^{-8} \text{ s}.$$

Why the difference. Both particles gain the *same kinetic energy*, $eEd = 1.6 \times 10^{-18} \text{ J}$. The proton is 1836 times heavier, so its speed is $\sqrt{1836} = 43$ times smaller, and it takes 43 times longer. The force on it is the same size, but in the opposite direction.

P13.9

Ring: from Example 13.4, $E = 2.5 \times 10^4 \text{ N/C}$.

Point charge:

$$E = \frac{kQ}{x^2} = \frac{(8.99 \times 10^9)(5.0 \times 10^{-8})}{0.12^2} = 3.1 \times 10^4 \text{ N/C}.$$

The ring's field is about 79% of the point charge's. Each element of the ring is farther away (13 cm instead of 12 cm), and only the axial component of its field counts. At greater distances the ratio approaches 1, because the ring then looks like a point.

P13.10

Field:

$$E = \frac{2k\lambda}{r} = \frac{2(8.99 \times 10^9)(2.0 \times 10^{-6})}{0.10} = 3.6 \times 10^5 \text{ N/C},$$

directed radially outward from the wire.

Force on an electron:

$$F = eE = (1.602 \times 10^{-19})(3.6 \times 10^5) = 5.8 \times 10^{-14} \text{ N},$$

directed towards the wire, since the electron's charge is negative.

P13.11

(a) **Dipole moment:**

$$p = qd = (2.0 \times 10^{-9})(4.0 \times 10^{-3}) = 8.0 \times 10^{-12} \text{ C m.}$$

(b) **Torque:**

$$\tau = pE \sin \theta = (8.0 \times 10^{-12})(5.0 \times 10^4) \sin 30^\circ = 2.0 \times 10^{-7} \text{ N m.}$$

(c) **Work from aligned to anti-aligned:**

$$W = U(180^\circ) - U(0^\circ) = pE - (-pE) = 2pE = 8.0 \times 10^{-7} \text{ J.}$$

P13.12

At **terminal speed**, the electric force balances the Stokes drag:

$$qE = 6\pi\eta r v \Rightarrow v = \frac{qE}{6\pi\eta r} = \frac{10(1.602 \times 10^{-19})(1000)}{6\pi(1.0 \times 10^{-3})(3.0 \times 10^{-9})} = \frac{1.60 \times 10^{-15}}{5.65 \times 10^{-11}} = 2.8 \times 10^{-5} \text{ m/s.}$$

In cm per hour: $2.8 \times 10^{-5} \times 100 \times 3600 = 10 \text{ cm/h.}$

Twice the radius: the drag doubles, so the speed **halves**, to about 5 cm/h. This dependence on size (and on charge) is what lets electrophoresis separate molecules. In a gel, the pore structure makes the separation by size even sharper.

P13.13

Put the rod along the x -axis, with linear charge density $\lambda = Q/L$.

(a) **On the axis.** Measure x from the field point, so the rod runs from $x = d$ to $x = d + L$. Every element's field points along the axis:

$$E = \int_d^{d+L} \frac{k\lambda dx}{x^2} = k\lambda \left[\frac{1}{d} - \frac{1}{d+L} \right] = \frac{k\lambda L}{d(d+L)} = \frac{kQ}{d(d+L)}.$$

(b) **On the perpendicular bisector**, at distance r . Measure x along the rod from its midpoint. The components along the rod cancel by symmetry. The perpendicular component from each element

is $\frac{k\lambda dx}{r^2 + x^2} \cdot \frac{r}{\sqrt{r^2 + x^2}}$:

$$E = k\lambda r \int_{-L/2}^{L/2} \frac{dx}{(r^2 + x^2)^{3/2}} = k\lambda r \left[\frac{x}{r^2 \sqrt{r^2 + x^2}} \right]_{-L/2}^{L/2} = \frac{k\lambda L}{r \sqrt{r^2 + L^2/4}} = \frac{2kQ}{r \sqrt{L^2 + 4r^2}}.$$

(c) Limits.

- In (a), with $d \gg L$: $E \rightarrow kQ/d^2$. ✓
- In (b), with $r \gg L$: $\sqrt{L^2 + 4r^2} \rightarrow 2r$, so $E \rightarrow kQ/r^2$. ✓
- In (b), with $L \rightarrow \infty$ at fixed λ : $\sqrt{r^2 + L^2/4} \rightarrow L/2$, so $E \rightarrow k\lambda L/(rL/2) = 2k\lambda/r$, the infinite-line result. ✓

P13.14

Rings. A ring of radius r' and width dr' carries charge $dq = \sigma 2\pi r' dr'$. On the axis it produces (Section 13.4)

$$dE = \frac{k dq x}{(x^2 + r'^2)^{3/2}}.$$

Integrate over r' from 0 to R :

$$E = 2\pi k\sigma x \int_0^R \frac{r' dr'}{(x^2 + r'^2)^{3/2}} = 2\pi k\sigma x \left[-\frac{1}{\sqrt{x^2 + r'^2}} \right]_0^R = 2\pi k\sigma \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right).$$

With $2\pi k = 1/(2\epsilon_0)$:

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right). \checkmark$$

Limits.

- Close to the disc ($x \ll R$): $x/\sqrt{x^2 + R^2} \rightarrow 0$, so $E \rightarrow \sigma/2\epsilon_0$, the infinite-sheet value. ✓
- Far away ($x \gg R$): use $\frac{x}{\sqrt{x^2 + R^2}} = \left(1 + \frac{R^2}{x^2} \right)^{-1/2} \approx 1 - \frac{R^2}{2x^2}$. Then

$$E \approx \frac{\sigma}{2\epsilon_0} \cdot \frac{R^2}{2x^2} = \frac{\sigma\pi R^2}{4\pi\epsilon_0 x^2} = \frac{kQ}{x^2}. \checkmark$$

P13.15

Set up. Put the centre at the origin, with the semicircle in the upper half-plane and the angle θ running from 0 to π . The linear density is $\lambda = Q/(\pi R)$. An element $dq = \lambda R d\theta$ at angle θ is a distance R from the centre, and its field there points away from the element, towards the centre and beyond.

Symmetry. The x -components from elements at θ and $\pi - \theta$ cancel. Every element's y -component points in the $-y$ direction, away from the arc.

Integrate:

$$E = \int_0^\pi \frac{k\lambda R d\theta}{R^2} \sin \theta = \frac{k\lambda}{R} \left[-\cos \theta \right]_0^\pi = \frac{2k\lambda}{R} = \frac{2kQ}{\pi R^2}.$$

The field points along the axis of symmetry, away from the midpoint of the arc (for positive Q).

Check. This is $2/\pi \approx 0.64$ of the field kQ/R^2 that would result if all the charge sat at the midpoint of the arc. It is smaller because the components from different parts of the arc partly cancel.

P13.16

(a) **Force on the electron.** On the axis at distance x , the ring's field is $E = kQx/(x^2 + R^2)^{3/2}$, pointing away from the centre. The electron's charge is negative, so it is pulled back towards the centre:

$$F = -\frac{kQex}{(x^2 + R^2)^{3/2}} \approx -\frac{kQe}{R^3}x \quad (x \ll R).$$

This is a linear restoring force, so the motion is SHM:

$$m\ddot{x} = -\frac{kQe}{R^3}x \Rightarrow \omega^2 = \frac{kQe}{mR^3}.$$

(b) **Numbers:**

$$\omega = \sqrt{\frac{(8.99 \times 10^9)(1.0 \times 10^{-9})(1.602 \times 10^{-19})}{(9.11 \times 10^{-31})(1.0 \times 10^{-6})}} = \sqrt{1.58 \times 10^{18}} = 1.26 \times 10^9 \text{ rad/s}.$$

$$f = \frac{\omega}{2\pi} = 2.0 \times 10^8 \text{ Hz} = 200 \text{ MHz}.$$

That is a radio frequency. Charges oscillating in fields like this are the basis of many sources of electromagnetic radiation.

P13.17

(a) **Field on the axis.** Put $+q$ at $x = +d/2$ and $-q$ at $x = -d/2$. At a point $x = r$ on the axis:

$$E = \frac{kq}{(r - d/2)^2} - \frac{kq}{(r + d/2)^2} = kq \frac{(r + d/2)^2 - (r - d/2)^2}{(r^2 - d^2/4)^2} = \frac{2kqrd}{(r^2 - d^2/4)^2}.$$

For $r \gg d$, the denominator is approximately r^4 , so

$$E \approx \frac{2kqd}{r^3} = \frac{2kp}{r^3}.$$

(b) **Water molecule at 1.0 nm:**

$$E = \frac{2(8.99 \times 10^9)(6.2 \times 10^{-30})}{(1.0 \times 10^{-9})^3} = 1.1 \times 10^8 \text{ N/C}.$$

Force on a Na^+ ion:

$$F = eE = (1.602 \times 10^{-19})(1.1 \times 10^8) = 1.8 \times 10^{-11} \text{ N}.$$

Force between the ion and a single electron at 1.0 nm:

$$F = \frac{ke^2}{r^2} = \frac{(8.99 \times 10^9)(1.602 \times 10^{-19})^2}{(1.0 \times 10^{-9})^2} = 2.3 \times 10^{-10} \text{ N}.$$

The dipole's force is about 13 times weaker, because of the $1/r^3$ cancellation. Even so, it is far larger than thermal forces at the molecular scale. That is why ions in water are tightly wrapped in oriented water molecules.

P13.18

(a) **Radius and mass.** At terminal speed with no field, the weight balances the Stokes drag:

$$\frac{4}{3}\pi r^3 \rho g = 6\pi \eta r v \Rightarrow r = \sqrt{\frac{9\eta v}{2\rho g}} = \sqrt{\frac{9(1.8 \times 10^{-5})(1.09 \times 10^{-4})}{2(900)(9.8)}} = \sqrt{1.0 \times 10^{-12}} = 1.0 \times 10^{-6} \text{ m}.$$

$$m = \frac{4}{3}\pi r^3 \rho = \frac{4}{3}\pi (1.0 \times 10^{-6})^3 (900) = 3.77 \times 10^{-15} \text{ kg}.$$

(b) **Charge.** With the drop held stationary, $qE = mg$:

$$q = \frac{mg}{E} = \frac{(3.77 \times 10^{-15})(9.8)}{7.68 \times 10^4} = 4.81 \times 10^{-19} \text{ C} = 3.0 e.$$

(c) **Quantisation.** Millikan (1909–1913) measured the charges on many drops, and on the same drop after it gained or lost charge by picking up ions from the air. Every charge he measured was a whole-number multiple of one smallest value, $e \approx 1.6 \times 10^{-19}$ C. Charges such as $2.5e$ never appeared. That showed directly that electric charge comes in discrete units.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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